Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience in **software verification**, **C++/Python systems**, and reliability-focused development using structured testing, simulation, traceable workflows, debugging, and technical documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Achieved 99%+ production reliability across 5K+ monthly jobs through **data validation**, reconciliation checks, error handling, cloud storage integration, and CI/CD releases connecting FastAPI services and relational databases.
- Tripled throughput across 1K+ AI jobs by restructuring **fault-tolerant orchestration** with explicit job states, failure isolation, recoverable processing, and observable workflows under concurrent production load reliably daily.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Maintained 90%+ grading consistency across 10K+ submissions by integrating **structured evaluation workflows**, rubric generation, model switching, fallback paths, persistent results, and instructor review controls for verification reliably.
- Processed 1K+ files per batch using **Python multithreading**, validating PDF, DOCX, and ZIP inputs while isolating failures and preserving repeatable workflows during concurrent model execution at scale.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Improved long-horizon trajectory planning by 18% in simulation by building an **algorithmic Python pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Raised closed-loop stability 15% through **quantitative verification** of HD maps, ego paths, nearby agents, attention, and distance-aware interactions across diverse autonomous-driving simulations and edge cases repeatedly and reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows at scale.
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments at scale.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship *India*

- Reduced centralized-control and quantum-risk exposure 25% in benchmark simulations by designing **consensus algorithms** and evaluating five lattice-based cryptographic primitives across distributed protocol scenarios for peer-reviewed research publication.
- Published peer-reviewed **systems research** and presented at IISc Bengaluru, documenting architecture, simulation methodology, threat models, experimental trade-offs, and reproducible technical findings for distributed consensus design and evaluation.

Skills

Languages: C++, Python, Java, SQL, Bash, Shell, JavaScript
Verification & Testing: Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Data Validation, Reconciliation Checks, Test Automation, Simulation, Debugging
Systems & Design: Linux, Object-Oriented Design, Data Structures, Algorithms, Concurrency, Multithreading, Distributed Systems, TCP/IP, System Design
Tools & Delivery: Git, GitHub, GitHub Actions, Jenkins, CI/CD, Docker, Kubernetes, AWS, Azure, GCP
Data & Practices: SQL Server, Performance Optimization, Technical Documentation, Research Methods

Projects

DO-178C Lifecycle Verification Package - Design Concept

- Designed a **DO-178C lifecycle verification package** concept covering planning, requirements traceability, review checklists, test procedures, expected results, structural-coverage records, change control, problem reports, and certification evidence organization.

Embedded Target Verification Bench - Design Concept

- Designed an **embedded C and Python verification bench** concept using requirement-based tests, hardware I/O stubs, boundary cases, fault injection, serial logs, coverage reporting, and repeatable regression execution.

CPU Scheduling Simulator | Live

- Implemented **C++ scheduling algorithms** for FCFS, SJF, SRT, Round Robin, priority, and MLQ, with edge-case handling, process-state visualization, and dynamic turnaround and waiting-time calculations across simulations reliably.